



DEALER BEST PRACTICE SERIES

Tracking Abusive Shifts on Large Off Highway Trucks

Application Maintenance Site Management Component Rebuild Component Life Management Safety MARC Management

Tracking Abusive Shifts on Large Off Highway Trucks 0

1.0 Introduction 1

2.0 Best Practice Description..... 1

3.0 Implementation Steps 2

4.0 Benefits..... 2

5.0 Resources Required 2

6.0 Supporting Attachments / References 3

7.0 Related Best Practices 7

8.0 Acknowledgements..... 7

DISCLAIMER: The information and potential benefits included in this document are based upon information provided by one or more Cat® dealers, and such dealer(s) opinion of “Best Practices”. Caterpillar makes no representation or warranty about the information contained in this document or the products referenced herein. Caterpillar welcomes additional “Best Practice” recommendations from our dealer network.

1.0 Introduction

Monitoring operator-induced events through VIMS is recommended to ensure machines are being adequately operated within their design parameters. Abusive shifts recorded by VIMS have been traditionally analyzed and reported as a single generic event.

Note that VIMS has the capability of reporting on the four different types of abusive shifts, these are:

- 1) First to reverse (directional shift)
- 2) Neutral to first
- 3) First to neutral
- 4) Reverse to first (directional shift).

Identifying the predominant 'type' of abusive shifts will help to implement corrective actions.

2.0 Best Practice Description

The following information outlines the process for logging an event due to an 'abusive' shift. An event will be logged with an engine speed of at least 1500 or 1800 rpm depending on the conditions and the truck model. Abusive shifting can result in reduced power train life and can be eliminated with operator training. Please note, the most recent software for 785, 789, 793 and 797 has been proven to not be a cause of abusive shifting characteristics.

- **First to reverse (Directional Shift)**

An event will be logged if the operator shifts the transmission control from first gear into reverse with an engine speed of at least 1,500 or 1,800 rpm (depending on conditions and truck model).

A transmission shift from first to reverse, whilst maintaining at least eight kilometres per hour (kph) ground speed, will hold the machine in forward gear. Once ground speed is less than eight kph the transmission will shift into neutral. Ground speed must be less than three kph before the transmission will shift into reverse.

- **Neutral to first**

An event will be logged when a shift occurs from neutral to first with an engine speed of at least 1,500 or 1,800 rpm (depending on conditions and truck model).

An abusive shift may occur by shifting into first gear when the engine speed exceeds 1,500 or 1,800 rpm (depending on conditions and truck model).

A practical scenario may be seen following a body raise whilst dumping.

The scenario is as follows:

Throttle is increased to induce power to hoist the body in a shorter time. Once the body is completely elevated, the hoist control lever is placed into the float position following a shift into first gear. Consequentially, there isn't time for engine speed to reduce to below 1,500 or 1,800 rpm (depending on conditions and truck model), resulting in a logged event.

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Tracking Abusive Shifts on Large Off Highway Trucks	DATE 2/16/2015	CHG NO 01	NUMBER 0911-2.04-1189
---	-------------------	-----------------	--------------------------

- **Neutral to Reverse**

An event will be logged when a shift occurs from neutral to reverse with an engine speed of at least 1,500 or 1,800 rpm (depending on conditions and truck model).

- **Reverse to first (Directional Shift)**

An event will be logged when the transmission shifts from reverse to forward with engine speed of at least 1,500 or 1,800 rpm (depending on conditions and truck model).

Please see section 6.0 'Supporting Attachments/References' for details pertaining to event limits.

Please see section 6.0 'Supporting Attachments/References' for details pertaining to the 793D Planetary Transmission, including clutch engagement information.

3.0 Implementation Steps

It is suggested that abusive shifts be tracked and trended over a period of time on a fleet of machines to identify any incorrect operating practices. Should these events be present, subsequent action will need to be taken by the onsite training department to eliminate and rectify any incorrect operating techniques.

4.0 Benefits

Benefits of analyzing and trending abusive shifts include but are not limited to:

- Improved machine operation
- Operator awareness and education
- Reduced downtime
- Optimized power train component life

5.0 Resources Required

The following items will be required to analyze the four different abusive shifts:

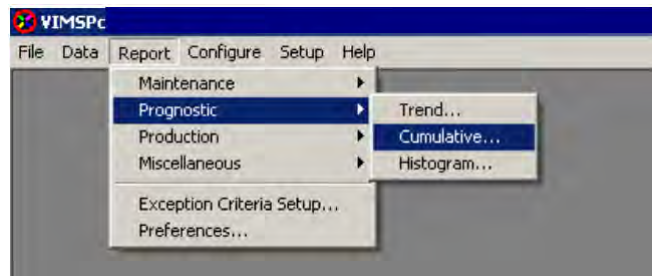
- VIMS equipped Off Highway Truck
- VIMS PC

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

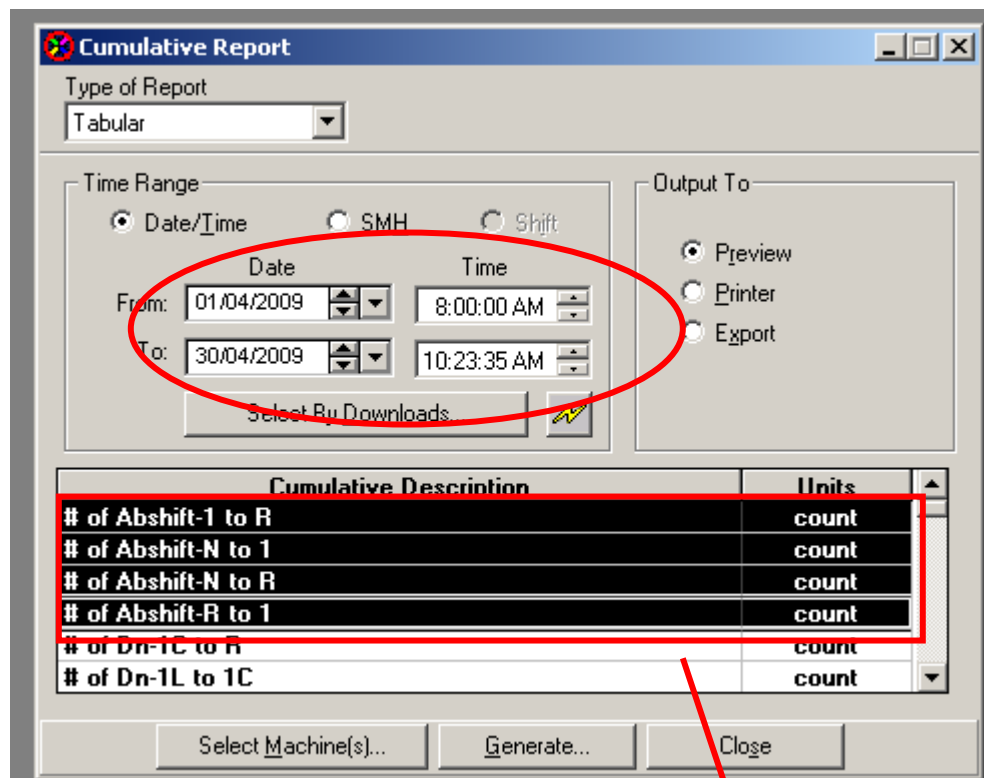
Tracking Abusive Shifts on Large Off Highway Trucks	DATE 2/16/2015	CHG NO 01	NUMBER 0911-2.04-1189
---	-------------------	-----------------	--------------------------

6.0 Supporting Attachments / References

Analyzing abusive shifts on VIMS PC



Select date range and select the four types of abusive shifts.



Printed: 27/05/2009 1:37:52PM

VIMSPc

Page 2 of 2

Cumulative Report

From: 1/04/2009 8:00:00 AM To: 30/04/2009 10:23:35 AM

Model Code:

Serial Number:

Equipment Number:

Cumulative Description	Units	Download Value at 28/04/2009 6:40:15 PM	Total (8/04/2009 6:05:06 AM - 28/04/2009 6:40:15 PM)
# of Abshift-1 to R	count	45	1
# of Abshift-N to 1	count	328	6
# of Abshift-N to R	count	34	0
# of Abshift-R to 1	count	20	0

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Tracking Abusive Shifts on Large Off Highway Trucks

DATE
2/16/2015

CHG
NO
01

NUMBER
0911-2.04-1189

Transmission shift event engine speed limits

797B	EOS to enable protection (not trigger event)	EOS to trigger event
Transmission	Engine Speed	Engine Speed
1F-R	1000	1500
R-1F	1000	1500
N-1F	1000	1800
N-R	1000	1800

Three second warning duration.

797F	EOS to enable protection (not trigger event)	EOS to trigger event
Transmission	Engine Speed	Engine Speed
1F-R	1000	1500
R-1F	1000	1500
N-1F	1000	1800
N-R	1000	1800

Ten second warning duration

785/789/793D	EOS to enable protection (not trigger event)	EOS to trigger event
Transmission	Engine Speed	Engine Speed
1F-R	1350	1500
R-1F	1350	1500
N-1F	1350	1500
N-R	1350	1500

Three second warning duration.

793F	EOS to enable protection (not trigger event)	EOS to trigger event
Transmission	Engine Speed	Engine Speed
1F-R	1350	1500
R-1F	1350	1500
N-1F	1350	1500
N-R	1350	1500

Ten second warning duration.

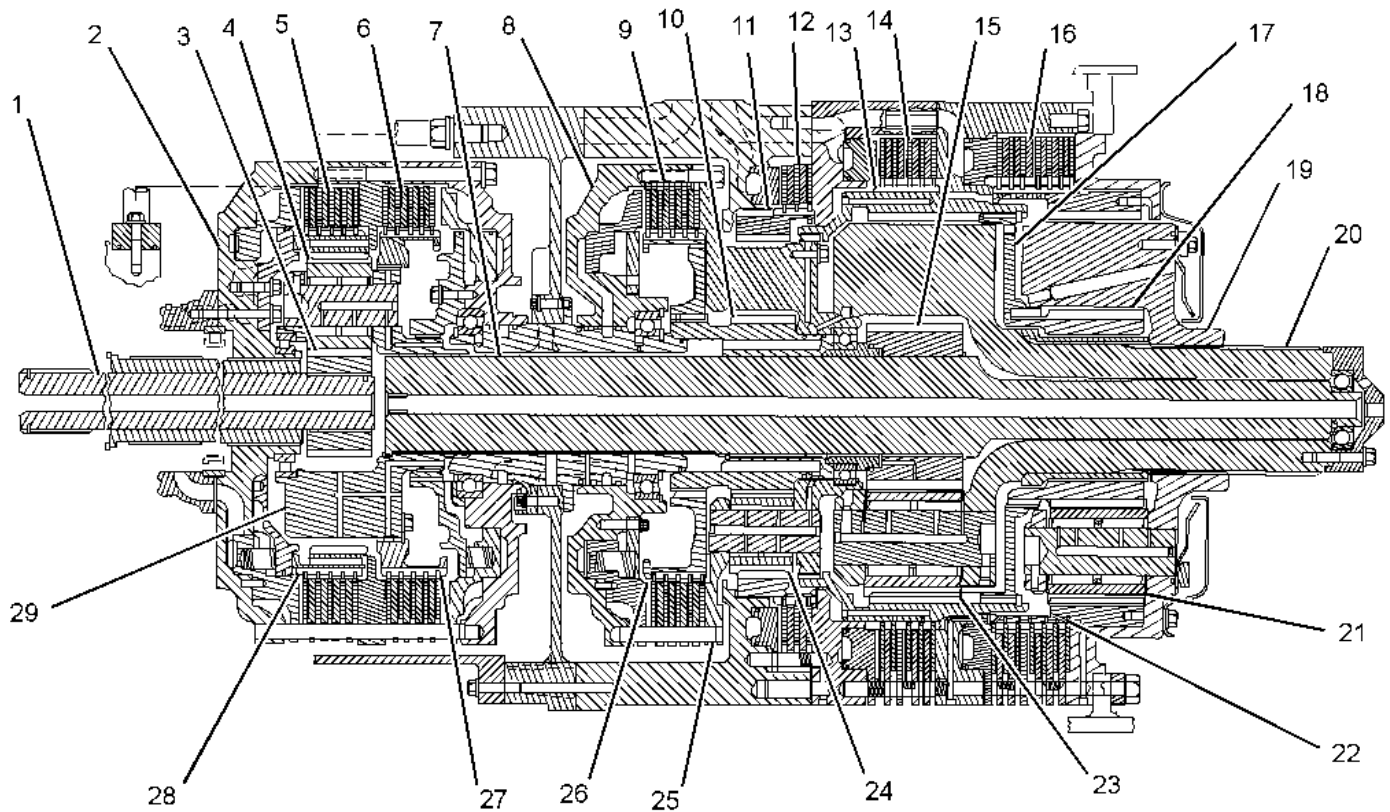
THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Tracking Abusive Shifts on Large Off Highway Trucks

DATE
2/16/2015

CHG
NO
01

NUMBER
0911-2.04-1189

793D Transmission – Planetary (REN8326 ‘Systems Operation – 793D OHT System Power Train - Transmission Planetary’)

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Tracking Abusive Shifts on Large Off Highway Trucks

DATE
2/16/2015CHG
NO
01NUMBER
0911-2.04-1189

Components of the transmission

(1) Input shaft	(16) No. 6 clutch
(2) Rotating housing	(17) Hub
(3) Sun gear	(18) Sun gear
(4) Planetary gears	(19) Planetary carrier
(5) No. 1 clutch	(20) Planetary carrier and output shaft
(6) No. 2 clutch	(21) Planetary gears
(7) Center shaft	(22) Ring gear
(8) Rotating housing of No. 3 clutch	(23) Planetary gears
(9) No. 3 clutch	(24) Planetary gears
(10) Sun gear	(25) Planetary carrier
(11) Ring gear	(26) Coupling hub
(12) No. 4 clutch	(27) Coupling ring gear
(13) Ring gear	(28) Ring gear
(14) No. 5 clutch	(29) Planetary carrier
(15) Sun gear	

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Tracking Abusive Shifts on Large Off Highway Trucks

DATE
2/16/2015CHG
NO
01NUMBER
0911-2.04-1189

Engagement of Transmission Clutches	
Transmission Speed	Engaged Clutches in the Transmission
NEUTRAL-1	3
NEUTRAL-2	1
REVERSE	1 and 6
FIRST speed	1 and 5
SECOND speed	2 and 5
THIRD speed	1 and 4
FOURTH speed	2 and 4
FIFTH speed	1 and 3
SIXTH speed	2 and 3

7.0 Related Best Practices

None

8.0 Acknowledgements

For further information regarding this Best Practice, please contact:

Rafael Castrillon
Development Engineer

Castrillon_Rafael@cat.com

Office +61 03 9953 9121

Cell +61 04 3745 1730

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Tracking Abusive Shifts on Large Off Highway Trucks	DATE 2/16/2015	CHG NO 01	NUMBER 0911-2.04-1189
---	-------------------	-----------------	--------------------------